Unpowered Walking Assistive Device Based on Gait Kinetic Energy Analysis

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***Abstract*—In recent years, there has been increased attention paid to the research of lower extremity walking assistive devices. However, many of them are powered by external energy and require separate actuators with corresponding control modules, which increase the overall quality while reducing the portability and wear-ability. In this paper, through the gait kinetic energy analysis, we summarized the movement rules of each joint in lower extremity to improve the walking efficiency. Considering that ankle joint whose movement rules are apparent has a significant influence on human gait performance, we proposed that there is a negative work stage in dorsiflexion, whereas plantarflexion does positive work by consuming muscle energy to drive the body forward. Therefore, we designed an unpowered walking assistant device, which imitated the normal gait by using the characteristics of energy storage under pressure of torsion spring in dorsiflexion and energy release in plantarflexion. The biodynamic simulation of the proposed design was conducted to verify the function of walking-assistant.**

Keywords—unpowered assistive device; kinetic energy analysis; gait; walking efficiency

I. INTRODUCTION

Human walking is the most basic and essential way of locomotion [1]. Although rapid transportation makes long-distance migration possible nowadays, walking becomes the best choice under the circumstances of weight-bearing and field scientific investigation. Various assistive devices have been proposed to augment the walking capability of human lower extremity recently, which can be mainly divided into two categories: fully powered and unpowered devices [2]. Powered walking assistive devices usually consist of external actuators, sensors and control modules, which increase the overall quality while reducing the portability and wear-ability. Compared with the powered ones, purely unpowered assistive devices possess merits of simpler mechanism, lighter weight, higher safety factor and stability control [3,4]. In this paper, through the analysis of normal gait, we demonstrated the timing of positive and negative work during gait and their corresponding lower extremity joint angles, which provides a theoretical basis for the design of the unpowered walking assistive device.

Choosing an appropriate energy storage and release strategy is the key factor to improve the efficiency of the unpowered walking assistive device [5]. Most of the energy consumption which driving the movement forward in a gait cycle comes from the elastic recoil of the Achilles tendon in the supporting phase [6]. With the aim of synchronizing the timing of the tendon recoil for energy release, many of the present unpowered components are designed to parallel position with the Achilles tendon [7-11]. However, the parallel arrangement increases the space occupied by the user's calf, which will bring mobility inconvenience to a certain extent. Mooney *et al.* [8] proposed a timing mechanism needs the cooperation of the foot pressure sensor and actuator to achieve energy storage and release, which is more complicated and requires additional power supply device. Collins *et al.* [9,10] used the ratchet mechanism to control the release of the spring energy, but the vibration of external environment may cause the pawl to break away from the moving range of the timing pin, causing the mechanism to fail. Kim *et al.* [11] adopted the control device with roller cam structure, which is designed according to the characteristics of energy storage and release in gait, this method makes the processing and installation process more complicated. Therefore, the key to optimizing the unpowered assistive device is to adopt appropriate energy storage components, flexible installation arrangement, and the simple and effective timing mechanism.

In this paper, we summarized the real-time angle and moment of lower extremity joints during gait and then validated the movement rules. On this basis, we designed an unpowered walking assistant device, which uses torsion spring as an energy storage element and installs it on the sole perpendicular to the Achilles tendon. Meanwhile, taking ratchet as the timing mechanism not only controls the accuracy and timely release of the stored energy but also locks the energy when the user is static, improved the stability and maneuverability. Lastly, we verified the energy efficiency performance by dynamic analysis.

II. GAIT EIGENVALUE ANALYSIS

A. Biomechanics of Gait

Before the kinetic energy analysis of human gait, we need to establish the lower extremity joint coordinate reference system. According to the anatomy of human lower extremity and the motion characteristics of corresponding joints, the reference frame of motion coordinates of the hip joint, knee joint and ankle joint is constructed [12]. Due to this study aims to analyze the gait of human forward walking, only the movement characteristics of the lower extremity joints in the sagittal plane are considered, that is, the extension and bending of the hip joint and the knee joint, the dorsiflexion and plantarflexion of the ankle joint, as shown in Fig. 1 (a). We also defined the angle direction of each joint on the sagittal plane based on their motion properties, as shown in Fig. 1 (b).

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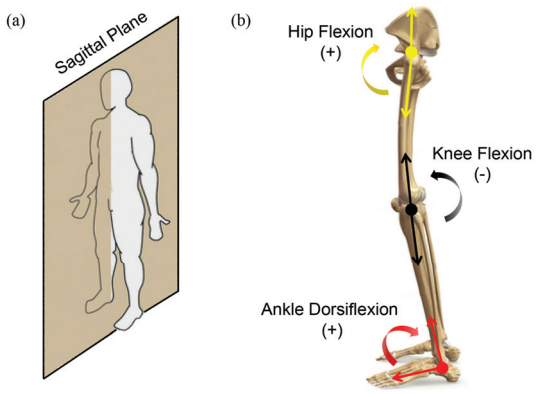

Fig. 1. (a) Sagittal plan; (b) angular direction of lower extremity joints.

Due to the human body follows specific walking patterns and rules, gait analysis can generally grasp the position, angle, acceleration, and moment of the joints. Gait cycle can be defined as the walking interval between two consecutive touches of the same heel, which can be divided into two stages: supporting phase and swing phase, as shown in Fig. 2. The supporting phase starts from the heel touch on one side to the end of the toe off the ground on the other side. The dorsiflexion of the ankle joint is the flat touch to the mid-stance, and the corresponding plantarflexion is the mid-stance to the propulsion. Current research only focuses on the supporting phase, because of the small motion range of the ankle joint in the swing phase.

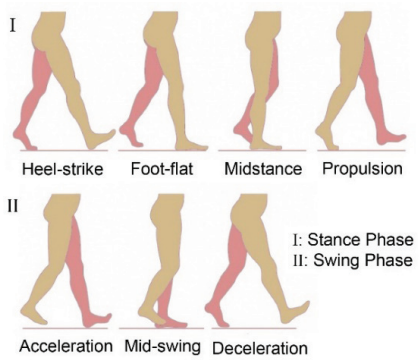

Fig. 2. Schematic diagram of the gait cycle.

B. *Time and Position of Positive and Negative Work in Gait*

To avoid the potential errors caused by different measuring devices and methods, we adopted the standard database that kinematics and dynamics samples of healthy humans walking on the flat ground [13]. The sample size was ten groups of adults, aged from 22 to 36, weighing about 68.4 kg and height about 171.9 cm.

Muscles and tendons together provide power for walking process and support for the body [14]. To study the force exerted on the lower extremity joints during gait, we plotted the above ten samples' moment diagrams of the hip, knee and ankle joints. As shown in Fig. 3(a), the moment provided by hip joints between different samples is significantly different and distributed over different gait periods. In contrast, as shown in Fig. 3(b), the characteristic distribution period of knee joints moment is relatively neat, but the initial moment between other samples differs significantly. However, ankle joints possess the almost same moment (two of which have a small range deviation in the initial stage), and the neat characteristic peak distribution, lighter noticeable difference as shown in Fig. 3(c).

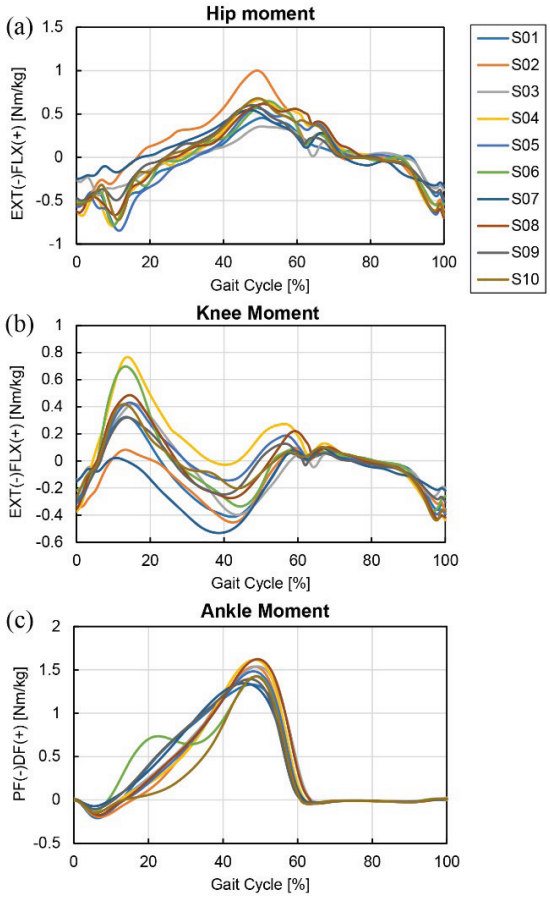

Fig. 3. Moment distribution of the hip, knee, and ankle joints during the gait cycle. The curves of the samples are the moment distribution of the hip and knee joint extension (EXT) and flexion (FLX), ankle joint plantarflexion (PF) and dorsiflexion (DF) in a single gait cycle.

To further understand the energy consumption of the lower extremity joints during gait, according to the actual motion state, we defined the distribution stage of the moment below the zero baseline as the muscle reactive power stage, due to muscle activity does not provide forward power for the walking process, which also can be depicted as negative work. On the contrary, we described the stage of moment distribution above the zero baselines as positive work, in which muscles provide kinetic energy for body forward. Although the three joints have positive and negative work phases as shown in Fig. 3, the work situation of hip and knee joint is distinct between different samples. The walking assistive devices based on these joints cannot provide comparison differentiated energy supplements. In the whole gait cycle, the distribution of positive and negative work of the ankle joint is apparent, and the moment provided between different samples is similar, there is no significant difference.

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Comparing with the differences of lower extremity joints during gait, we selected the average real-time angle distribution of each joint of the above samples, as shown in Fig. 4. The walking angle of hip joint walking angle is between 0° and 30°, that of the knee joint is between -1° and 60°, and that of the ankle is between -12° and 15°.

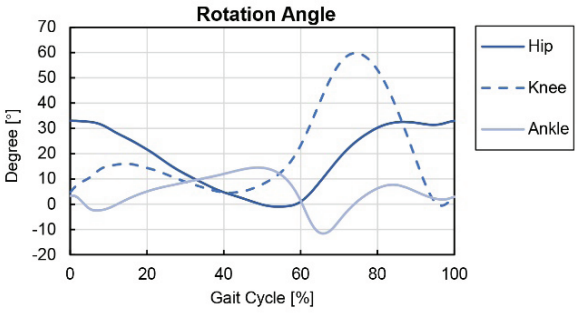


Fig. 4. Walking angle distribution of hip, knee, and ankle during gait.

By virtue of the analysis above, we found that the walking angle span of the knee joint is relatively large, whereas that of hip and ankle joint are about 30°. The overall trend of hip and ankle joint are relatively gentle, indicating that there is no more rapid acceleration change in motion. Based on the physiological structure of the lower extremity joints, the hip joint possesses high flexibility, and its movement is stable but complex; knee joint is affected by hip and ankle joint movement, and easy to tremble; the ankle joint transmits the supporting force to the leg bone while ensuring the stability of the walking gait. Therefore, the further experimental verification focuses on the analysis and verification of ankle joint and summarizes the distribution stage of positive and negative work during gait.

III. Data Acquisition and Analysis

To thoroughly study the motion rules of the ankle joint in a gait cycle, and to ensure the accuracy of gait motion information acquisition, we used a wearable sensor device with high precision and portability. The obtained experimental data was from the Motion Data Acquisition and Analysis System (myoMOTION Software Module; Noraxon, Inc., USA), and the foot pressure insole (Medilogic Insoles; Noraxon, Inc., USA).

Five subjects (male, 23.6±1.6yrs, height 172.0±3.1cm, weight 71.1±2.2kg) were selected for the kinetic energy analysis experiment of ankle joint. We collected the real-time angle, acceleration and vertical ground support force. Averaged experimental data obtained from subjects' comfortable walking speed on the ground and the walking time was rounded to a gait cycle. We extracted the stage signs of the ankle joint in the supporting phase: foot-flat and mid-stance, which correspond to the starting points of the dorsiflexion and plantarflexion, respectively, as shown in Fig. 5.

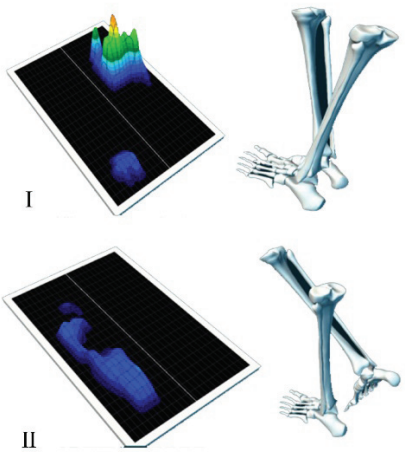
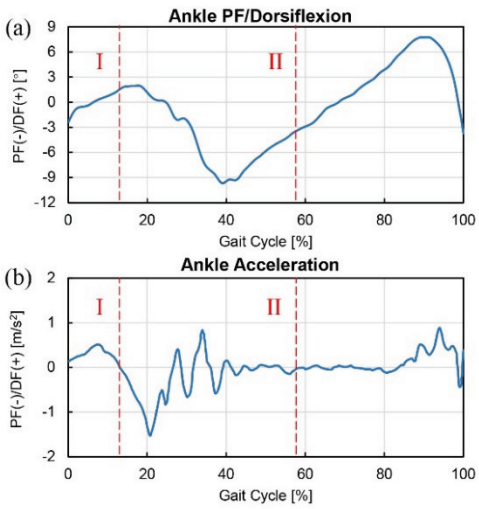


Fig. 5. Initial movement of ankle dorsiflexion (I) and plantarflexion (II) and the plantar pressure distribution.

Fig. 5(I) shows the initial period of the gait measured, and corresponds to Fig. 6-I (the left red dotted line), i.e., foot-flat in dorsiflexion at the largest angle with the calf. Acceleration of left foot gradually changes from positive to negative, and the velocity direction is positive and negative work begins. Afterward, the ankle joints slowly entered the initial stage of plantarflexion. Fig. 5(II) shows the middle period which corresponding to Fig. 6-II (the right red dotted line), that is, the foot is perpendicular to the calf (defined as 0°). Acceleration of left foot gradually changes from negative to positive, the velocity direction is positive and positive work begins.

Combined with the above comprehensive analysis of movement angle and plantar pressure distribution, we get the initial position of dorsiflexion and plantarflexion. Lower extremity gradually turns from negative power consumption to positive power output, thus pushing the body forward. Therefore, we propose the structure of the unpowered walking assistive device, and then simulate it to verified the efficiency performance by dynamic analysis.



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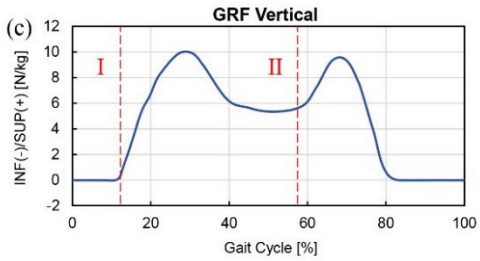

Fig. 6. Ankle joint parameter curves. The measured curves were the average values of real-time angle (a), acceleration (b) and ground reaction force (c) in the ankle joint. The data are all the variation curves of medial ankle plantarflexion (PF) and dorsiflexion (DF), and the range of GRF is expressed as the value between infimum (INF) and supremum (SUP).

IV. Modeling and Simulation

A. Modeling of Unpowered Walking Assistive Device

For structure of the unpowered walking assistive device, the aim is to collect the negative work of ankle dorsiflexion and release it in plantarflexion. The device is mainly composed of ratchet mechanism and energy storage torsion spring. As shown in Fig. 7, its components connect the front and rear parts of the device, allowing the dorsiflexion and plantarflexion of the ankle joint. The ratchet mechanism controlled by linear spring meets the need of braking device at any time. At the same time, the release of torsion spring energy can reduce the working moment during gait.

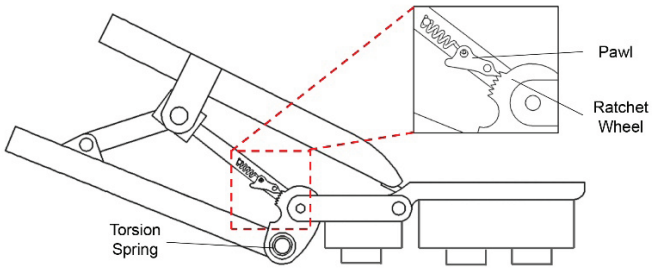

Fig. 7. Schematic diagram of the unpowered walking assistive device.

The anti-backlash pawl is above the gear teeth in support phase, and the torsion spring is compressed with the angle of the device to store energy until the entire sole touches the ground. If the human body remains stationary, the anti-backlash pawl engages with the gear teeth, thus hindering the upward trend, as shown in Fig. 8 (b); if continues to move, slightly lower pressure is needed to make the anti-backlash pawl cross the chute and release the meshing state, as shown in Fig. 8 (c); if keeps moving, the inherent inertia of the anti-backlash pawl will automatically cross the chute while the pre-compressed torsion spring releases to assist locomotion.

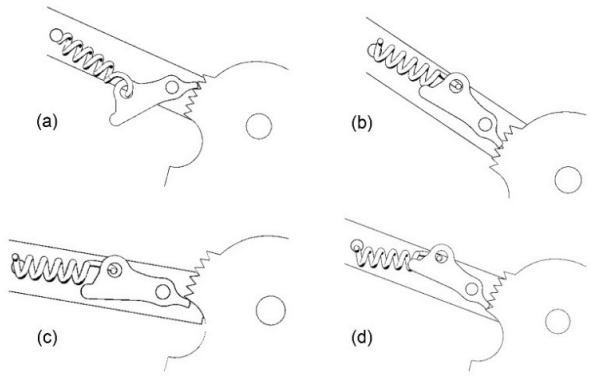

Fig. 8. Ratchet working mode of the in unpowered walking assistive device: (a) operating; (b) backstop; (c) release; (d) upward.

B. Biomechanical Energy Efficiency Simulation Analysis

According to the function layout of the torsion spring mentioned above, we use the method of simplifying the torsion spring into the ankle joint assistant force to simulate the biodynamics. The three-dimensional walking human model we generated is a musculoskeletal simulation dynamic model driven by 92 muscles and tendons [15] as shown in Fig. 9, and the gait data is collected from the normal walking speed of the real human. Simulation step is to compare the working moment before and after adding the assistant torsion spring to verify the auxiliary efficiency, so as to obtain the design rationality of the assistive device.

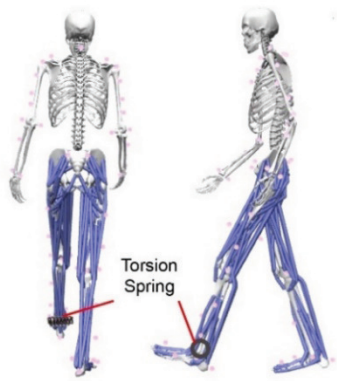

Fig. 9. Three-dimensional muscle-driven walking human model with torsion springs.

We used two standard gait samples in this section, simulation and data calculation were performed by OpenSim 4.0 [16], which is an open source software for biomechanical modeling, simulation, and analysis. After adjusting the appropriate parameters and action intervals of the torsion spring, the sample is measured and analyzed. It is concluded that the torsion spring is compressed with negative moment, then released gradually in plantarflexion, and the peak moment value decreases between 5.6% and 9.5%. The size of the overall auxiliary work is the same as the energy of the negative work phase of the dorsiflexion, as shown in Fig. 10.

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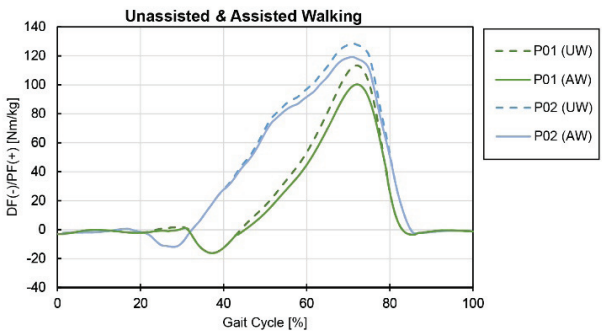

Fig. 10. Schematic diagram of the working moment of the ankle joint assisted by the torsion spring (UW: unsisted by torsion spring; AW: assisted walking).

V. Conclusions

In this paper, we summarized the movement rules of the hip, knee and ankle joints according to the gait data from the regular human walking, in which ankle joint has a more significant impact on gait. Cooperate with the real-time rotation angle and working moment of the ankle joint, we proposed that there is a waste of energy in dorsiflexion, while forward movement in plantarflexion causes a higher energy consumption. We designed an unpowered walking assistive device, which uses a torsion spring to store energy compressed in dorsiflexion, release the energy in plantarflexion imitating the normal gait, and then verified the efficiency performance by dynamic analysis.

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